

MERN Task Manager

Full-Stack Task Management Application

A full-stack task management application built with the MERN stack. It supports user authentication, personal task management, filtering, search, pagination, dark mode, and a demo admin dashboard.

Overview

MERN Task Manager is designed as a practical CRUD application for organizing tasks. Users can sign up, log in, create tasks, update progress, set priorities and due dates, search through their tasks, and manage task status from a React dashboard. The backend provides a protected Express API with MongoDB persistence and JWT-based authentication.

Tech Stack

- MongoDB with Mongoose
- Express.js
- React 18 with Vite
- Node.js
- JWT Authentication
- Axios
- Docker Compose

Features

- User signup and login
- JWT-protected API routes
- Create, read, update, and delete tasks
- Task status tracking: **todo**, **in-progress**, and **completed**
- Task priority levels: **low**, **medium**, and **high**
- Optional task descriptions and due dates
- Search tasks by text
- Filter tasks by status
- Paginated task list
- Dashboard task statistics
- Dark mode support
- Protected frontend routes
- Demo admin dashboard
- Admin task visibility and delete capability
- Centralized API error handling
- Docker setup for client, server, and MongoDB

Demo Login

Email: admin@demo.com
Password: admin123

Admin Dashboard:
<http://localhost:5173/admin>

Environment Setup

```
copy server\.env.example server\.env  
copy client\.env.example client\.env
```

```
PowerShell:  
Copy-Item server\.env.example server\.env  
Copy-Item client\.env.example client\.env
```

Local Setup

```
cd server
npm install

cd ../client
npm install

# Start backend
cd server
npm run dev

# Start frontend
cd client
npm run dev
```

Docker

```
docker compose up --build
```

API Routes

```
Auth:
POST  /api/auth/signup
POST  /api/auth/login
GET    /api/auth/me

Tasks:
GET    /api/tasks
POST   /api/tasks
PATCH /api/tasks/:id
DELETE /api/tasks/:id

Admin:
GET    /api/admin/tasks
DELETE /api/admin/tasks/:id
```

Project Structure

```
mern-task-manager/
├── client/           React + Vite frontend
├── server/           Express + MongoDB backend
├── docker-compose.yml
└── README.md
```

Future Improvements

- Add TypeScript across the client and server
- Add form validation with clearer field-level messages
- Add automated tests for API routes and React components
- Add task labels or categories
- Add task sorting by due date, priority, and creation date
- Add password reset and email verification
- Add refresh tokens for stronger authentication flow
- Improve analytics on the dashboard with charts and task trends